

Advanced Subnetting Techniques & Problem Solving (IPv4)

Focus

Subnetting dominates IPv4 questions in **GATE (CS/IT)** because it combines:

- Binary logic
- Mathematical reasoning
- Configuration awareness

This briefing shifts the mindset from **slow binary expansion** to **logical bit-weight analysis and shortcuts**, especially the **AD Rule**, enabling faster and more reliable answers under exam pressure.

1 Fundamentals of Subnet Masking (Reinforced Logic)

A subnet mask is nothing more than a **bit-distribution blueprint**.

Core Interpretation Rules

- **Binary 1s (Mask)** → Network ID + Subnet ID
- **Binary 0s (Mask)** → Host ID

Mathematical Relationships

- Total **1s** = Network bits + Subnet bits
- Total **0s** = Host bits

Host Capacity

If Host ID bits = **n**:

- Total IPs per subnet = 2^n
- Usable hosts = $2^n - 2$

(Subnet ID + Directed Broadcast Address are excluded)

Default Class Structures (Baseline Reference)

Class	Network Bits	Host Bits
A	8	24
B	16	16
C	24	8

These defaults are **only a starting point**; subnetting modifies them.

2 Requirements-Driven Subnetting (Exam First Principle)

In real problems, **requirements dictate the mask**, not the class.

◆ Determining Host ID Bits (Host-Driven)

To support **H hosts**:

Find smallest **n** such that:

$$\begin{aligned} [& \\ 2^n - 2 & \geq H \\] \end{aligned}$$

Example – 100 Hosts

- 6 bits $\rightarrow 2^6 - 2 = 62$ ❌
- 7 bits $\rightarrow 2^7 - 2 = 126$ ✅

✓ Host ID = 7 bits

Class B Impact

- Default Network bits = 16
- Remaining bits = $16 - 7 = 9$ (Subnet bits)
- Total 1s = $16 + 9 = 25$

👉 Mask = 255.255.255.128 (/25)

♦ Determining Subnet ID Bits (Subnet-Driven)

To support **S** subnets:

Find smallest **n** such that:

$$\begin{bmatrix} 2^n \geq S \\ \end{bmatrix}$$

Example – 150 Subnets

- 7 bits → 128 ❌
- 8 bits → 256 ✅

✓ Subnet bits = 8

Class B Impact

- Host bits left = 16 – 8 = 8
- Total 1s = 16 + 8 = 24

👉 Mask = 255.255.255.0 (/24)

3 AD Rule – High-Speed Subnet & Host ID Detection

The **AD Rule** eliminates full binary conversion.

♦ Finding Subnet ID (Fast Method)

Subnet ID = IP AND Mask

Shortcut Rules

- Mask octet = **255** → Subnet octet = IP octet
- Mask octet = **0** → Subnet octet = 0
- Partial mask (e.g., 192, 224, 240) → analyze only **that octet**

Partial Octet Trick

Instead of full binary:

- Write bit weights: **128 64 32 16 8 4 2 1**
- Add only weights common to **IP octet & mask**

♦ Finding Host ID (Instantly)

No binary required.

```
[  
\text{Host ID} = \text{IP Address} - \text{Subnet ID}  
]
```

Example

- IP last octet = 200
- Subnet ID octet = 192

👉 Host ID = 8

This works because:

- Subnet ID zeroes host bits
- Remaining value = host portion

4 Directed Broadcast Address (DBA) Analysis

A **DBA** is defined as:

All Host ID bits = 1

♦ Validating a DBA Using a Mask

If Host ID bits = **n**:

- The IP address **must end with n binary 1s**

Example

- Host bits = 4
- Valid DBA must end with: **....1111**

♦ Finding Possible Masks from a Given DBA (Advanced GATE Trick)

Steps:

1. Convert DBA's last octet to binary
2. Count **continuous trailing 1s from right**
3. That count = **maximum possible Host ID bits**

Rule

[
 $\text{Mask Host Bits} \leq \text{Trailing 1s in DBA}$
]

Example

- DBA last octet = **31** → **00011111**
- Trailing 1s = 5

✓ Valid masks must have **≤ 5 Host bits**

5 Core GATE Problem Patterns (Must-Master)

Question Type

Fast Logic

Number of subnets	Extra 1s beyond default → 2^{extra}
IPs per subnet	Count 0s in mask → 2^{zeros}
Subnet ID	AND logic or bit-weight overlap
DBA validation	Host portion must be all 1s

6 High-Yield Observations

- ✓ One mask can apply to **Class A/B/C**, but subnet counts differ
- ✓ Subnetting mastery alone covers ~80% of IPv4 GATE questions
- ✓ Non-continuous masks are **theoretical traps** testing AND logic
- ✓ Real networking uses **continuous 1s**, exams may not

